

THE GREAT BEAR

Ursa Major



SIZE RANKING 3

BRIGHTEST STARS

Alpha (α) 1.8,
Epsilon (ϵ) 1.8.

GENITIVE Ursa
Majoris

ABBREVIATION UMa

HIGHEST IN SKY AT 10 P.M.
February–May

FULLY VISIBLE
90°N–16°S



Ursa Major is one of the best-known constellations and a prominent feature of the northern sky. Seven of its stars form the familiar shape of the Big Dipper. But as a whole, Ursa Major is much larger than this; it is the third-largest constellation in the sky. The two stars in the Big Dipper's bowl farthest from the handle, Dubhe (α) and Merak (β), point toward the north Pole Star, Polaris, while the curved handle of

the Big Dipper points toward the bright star Arcturus in the adjoining constellation of Boötes.

SPECIFIC FEATURES

The Big Dipper is one of the most famous patterns in the sky. Its shape is formed by the stars Dubhe (α), Merak (β), Phecda (γ), Megrez (δ) Ursa Majoris, Alioth (ϵ), Mizar (ζ) (see p.276), and Alkaid (η). With the exception of Dubhe and Alkaid, these stars travel through space in the same direction, and they form what is known as a moving cluster.

Mizar (ζ), the second star in the Big Dipper's handle, is next to Alcor (see p.276), an eighth, fainter star in the Big Dipper, which can be seen with good eyesight. A small telescope reveals that Mizar also has a closer 4th-magnitude companion.

In southern Ursa Major lies another attractive double star, Xi (ξ) Ursa Majoris, which can easily be divided through only a small telescope. This pair, with components of 4th and 5th magnitudes, forms a true binary

relationship, orbiting every 60 years, which is quick by the standards of visual binary stars.

One of the easiest galaxies to identify with binoculars is M81, which is in northern Ursa Major, and is also known as Bode's Galaxy (see p.314). This spiral galaxy is at an angle and can be seen on clear, dark nights as a slightly elongated patch of light. A telescope is needed to spot the rather more elongated shape of the smaller and fainter Cigar Galaxy (see p.314), or M82, which is found one diameter of a full moon away from Bode's Galaxy. This unusual-looking object is now thought to be a spiral galaxy, seen edge-on, mottled with dust clouds and undergoing a burst of star formation following an encounter with M81.

Another major spiral galaxy in this constellation is the Pinwheel Galaxy, M101 (p.316), which lies near the end of the Big Dipper's handle. Though larger than Bode's Galaxy, it is fainter and thus more difficult to see. An even greater challenge to find and



THE OWL NEBULA ☾

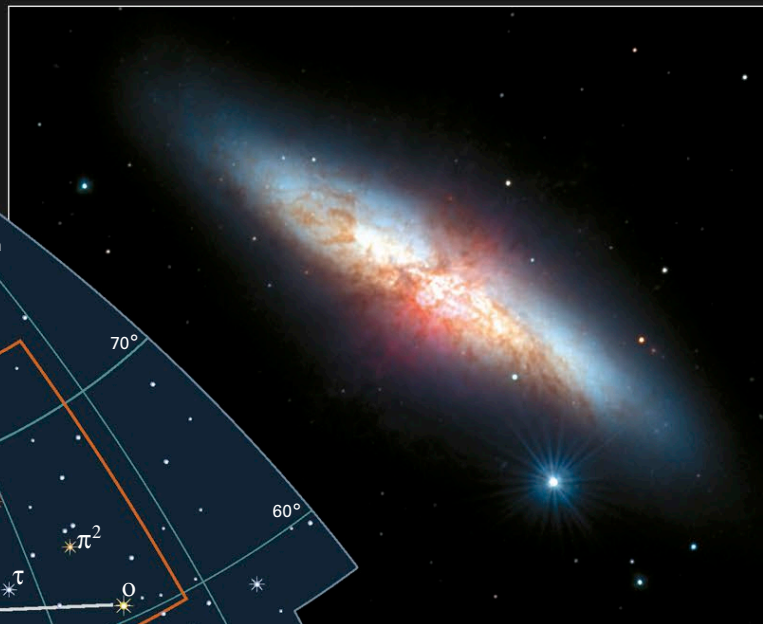
The dark, owl-like eyes of the faint planetary nebula M97 are visible only through large telescopes or on photographs and CCD images such as this one.

identify, however, is the Owl Nebula, or M97, located under the bowl of the Big Dipper. This planetary nebula is one of the faintest objects in Charles Messier's catalog, and a telescope of around 3 in (75 mm) aperture is needed to make out its gray-green disk, which is three times larger than that of Jupiter. A telescope with an even larger aperture reveals the two dark patches, like an owl's eyes, that give rise to its popular name.

THE CIGAR GALAXY ☾

M82 is a peculiar-looking spiral galaxy, edge-on to us, that is undergoing a burst of star formation triggered by a close encounter with the larger and brighter spiral galaxy M81 about 300 million years ago.

THE GREAT BEAR



BODE'S GALAXY ☾

This spiral galaxy was discovered by German astronomer Johann Elert Bode on December 31, 1774. Located approximately 11 million light-years away, M81 is nevertheless one of the brightest and most visible galaxies in the sky.